

Pooja Ravi

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EDUCATION

MSc, Applied Computing

2023-2025

University of Toronto, Canada

Awards: Vector Scholarship in Artificial Intelligence (2023-24) | Mitacs Globalink Graduate Fellowship

BTech, Computer Science and Engineering

2019-2023

SRM Institute of Science and Technology, India

RESEARCH EXPERIENCE

Researcher | University of Toronto (TCIG)

Apr 2026 - Present

Supervised by David B. Lindell

In-hand Robotic Object Manipulation

- **Action recovery:** Tracked ray-traced hand-link, palm, and wrist query points with CoTracker; solved frame-wise inverse kinematics by minimizing reprojection error, warm-started from previous-frame angles. Refined target joint angles via constrained cross-entropy method (CEM) search to improve the intended manipulation motion.
- **Physics replay:** Executed recovered actions in MuJoCo with contact and gravity; passed a frame from each replay back as the next video-generation cycle's initial frame, closing the loop.
- **Closed-loop pipeline:** Integrated a fine-tuned object-motion-conditioned video model with CoTracker, inverse kinematics, and constrained CEM refinement for ORCA Hand v1, connecting generated video to refined joint-space actions and MuJoCo feedback.

ProxyPose (NeurIPS 2026)

- **Benchmarking:** Engineered a unified 6-DoF benchmark across HO3D, YCBinEOAT, and YCB-Video; standardized meshes, RGB/depth frames, pose annotations, and metadata in a consistent evaluation format.
- **Evaluation:** Implemented task-aligned metrics and dataset-specific evaluation pipelines; integrated competing methods, ran cross-dataset experiments, and analyzed comparative results.

PUBLICATIONS

1. ProxyPose: 6-DoF Pose Tracking via Video-to-Video Translation

Ruihang Zhang, Felix Taubner, Pooja Ravi, Kiriakos N. Kutulakos, and David B. Lindell

NeurIPS · 2026

2. Attention Mechanism, Linked Networks, and Pyramid Pooling Enabled 3D Biomedical Image Segmentation

Pooja Ravi, Srijarko Roy, Indira Dutta, and Kottilingam Kottursamy

IEEE/ACIS SNPD · 2022

3. Real-time Multi-Module Student Engagement Detection System

Pooja Ravi and Ali Dewan

ICCIS · 2022

INDUSTRY EXPERIENCE

Machine Learning Engineer | Q2 Software

May 2024 - Mar 2026

- Led cheque-signature verification across 200K images, training variational autoencoder embeddings and Siamese networks with triplet and contrastive losses; fine-tuned multimodal LLMs for pairwise visual comparison.
- Trained LayoutLM, Donut (Swin+BART), and TrOCR on 100K+ cheques; improved text extraction by 8% and fraud flagging by 4% while reducing inference latency through model optimization.

Machine Learning Engineer | 16 Bit

Oct 2023 - Mar 2024

- Led temporal X-ray image-to-image translation between baseline and follow-up scans (12-96 months) to study knee osteoarthritis progression; scaled GAN and diffusion experiments with PyTorch Lightning/LoRA on 70K+ images, achieving ~10x faster training and ~19% higher image realism.

SELECTED PROJECTS

Echo Charlie: multimodal voice restoration system; Boson AI Hackathon winner (2025).

2D Sketch to 3D Stylization: NeRF-based sketch stylization through image style transfer and CLIP-guided text prompts.

MobileGen: optimized DDPM for mobile inference using quantization-aware training, distillation, and pruning; 20% faster with 14% fewer GFLOPs.